

Product Information Cleaning and Transformation

Objective:

This project is created for cleaning and transforming product related information in a structured format. The goal is for handling missing values, correcting inconsistencies, removing duplicate records, restructuring columns, and applying formatting rules using Excel for performing meaningful analysis.

Dataset Description:

The dataset contains the following columns.

Column description:

Column Name	Description
Product ID	Number of the Product
Product Name	Name of the Product
Brand Name	Brand Name of the Product
Price	Price of the Product
Quantity	Quantity of the Product
Category	Product's Category
Manufacturing Date	Newly extracted "Manufacturing Date" from Product ID
Country Code	Newly extracted "Country Code" from Product ID
Product Brand	Newly Merged the Brand and Product Name

Formulas Used:

Handling Missing Values

A. How would you handle products with missing price information

Answer:

Missing values can be replaced as "Average Price" Hihglighted in "BOLD"

Formula: =IF(ISBLANK(G2),AVERAGE(G2:G32),G2)

B. Products with missing categories, propose a strategy

Answer:

Missing Category filled with the text "Unknown" highlighted in "BOLD"

Formula: =IF(ISBLANK(I2),"Unknown",I2)

Or

Find Mode function to replace blank cell with highest repeated names in Category column

Formula:

=IF(ISBLANK(I2),INDEX(I2:I32,MODE(MATCH(I2:I32,I2:I32,0))),I2)

Correcting Inconsistent Data

A. Identify any inconsistent text formats present in the "Product Name" column

Answer:

Yes, Changed all name to "Proper case"

Formula: =PROPER(D2:D32)

B. Identify any typos present in the "Category" column

Answer:

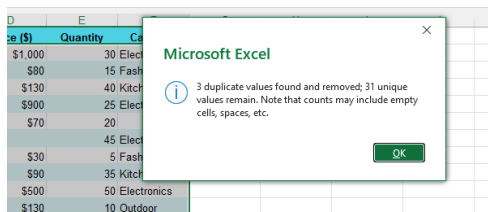
Yes, Electroni changed to Electronics using "Find and Replace" option

Removing Duplicates

A. Identify any duplicate rows within the dataset and remove them if any

Answer:

Yes, 3 duplicate values were found in "Product ID" column and removed using "Remove Duplicates" option



Splitting and Merging Data

A. Split the "Product ID" column into two separate columns

Answer:

One column Splitted as "Country Code" using "Text to Columns" option

Another column splitted as "Manufacturing Date" using follwing formula

Formula: =C2&"-"&D2

B. Merge the "Brand Name" and "Product Name" columns into "Product Brand"

Answer:

Yes, Merged into as "Product Brand"

Formula: =E2&" "&D2

Number Formatting

A. Format the data type of the "Price" column to a currency format

Answer:

Yes, Price column changed from "General" format to "Currency" format

B. Format the "Manufacturing Date" column to "DD-MM-YYYY" format

Answer:

Yes, Changed all date format to "28-Jan-2026" format

Conditional Formatting

A. Apply conditional formatting in the "Price" column

Answer:

Yes, Applied the "Data Bar" in the price column highted in PINK colour

B. Create a custom rule for conditional formatting in the "Category" column of "Electronics"

Answer:

Yes, highlighted in YELLOW colour

Conclusion:

This project is help to develop practical skills in data preprocessing, standardization, and quality improvement. It enhances data analysis through data cleaning techniques, formatting tools, and transformation operations.

